GRASP: Gated Regression-Aware Skill Proposer for Self-Improving LLM Agents

Johannes Moll, Jean-Philippe Corbeil, Jiazhen Pan, Martin Hadamitzky, Daniel Rueckert, Lisa Adams, Keno Bressem

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Problem

Self-improving agents add notes from their failures but rarely check whether a new note **breaks cases already solved**.

> *A note that fixes one trajectory can silently regress another.*

Motivation

- To improve **monotonically**, each change needs a causal test against what already works.

In the paper, Sequential and Batch memory fall below the no-skills baseline as memory accumulates.

Our read · HarnessRSI-Daily

84/100 ★★★ · evidence level **retained**

REPORTED	NOT / PARTLY
Held-out test + OOD	Leakage test
3–5 seeds, ± s.d.	Matched compute (partial)
Call cost · code	

Changes its Markdown skill library only (no weights) · **signal**: its own failing and passing traces · **kept** only through the probe gate.

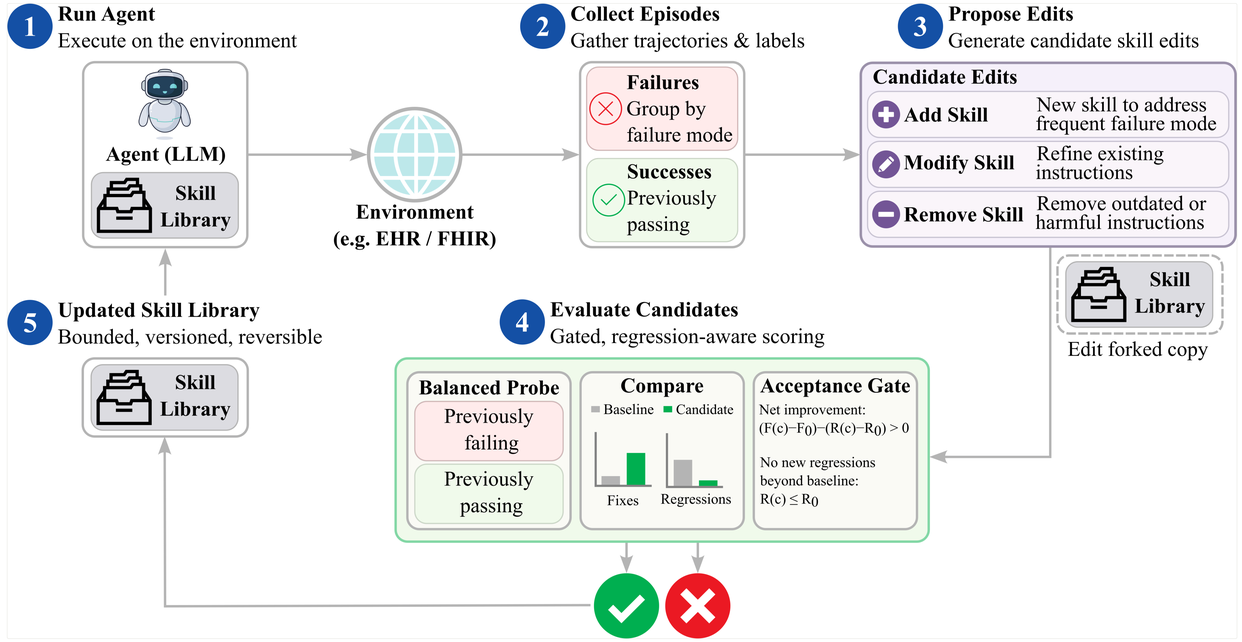
Gains concentrate on MedAgentBench; FHIR-AgentBench is level with baselines, and OOD spread is wide (56.3 ± 14.5).

Method

1 **Propose** K edits (ADD / MODIFY / REMOVE, ≤10 skills) from grouped failures.

2 **Probe** each on a forked library: half failing, half passing dev cases.

3 **Gate**: keep the best only if $(F_c - F_0) - (R_c - R_0) > 0$ and $R_c \le R_0$ (F fixes, R regressions vs. the current library).



Overview figure from the authors' repository github.com/jomoll/GRASP (MIT); paper CC BY 4.0.

Ablation Study

NO GATE	vs.	GATE KEPT
Apply every edit: 40.1% (K=1), 63.5% (K=4).		Other parts removed: 80–84% (5-epoch budget).

SO WHAT → *Skill writing without validation is no better than no skills; the gate carries the gain.*

- Frozen libraries transfer across models and benchmarks that share a tool-calling convention, and degrade under interface mismatch.

Key Results



MedAgentBench test acc. (%), gpt-oss-120b, 5 seeds. Drawn by us from the authors' released results.

+21.0 pts over Evo-MedAgent, the best baseline; OOD 56.3% vs 31.3%.

Dataset / Benchmark

MedAgentBench · MedAgentBench-v2 · FHIR-AgentBench · ALFWorld · WebShop · DBBench · OS

- Five base models, 3–5 seeds each; test and OOD splits never used in training.

Headline Numbers

88.8%
MedAgentBench test · gpt-oss-120b
from 40.6% with no skills (5 seeds)

+21.0 pts over best baseline
40.1% gate removed (K=1)
3/4 non-clinical envs improved

Takeaway

A self-written skill library helps reliably only when each edit must beat the current library on a balanced held-out probe.

The cost is extra training-time calls for validation, almost all from probing. Libraries written by a stronger model help weaker ones; the reverse does not hold.

Outside medicine: ALFWorld 23.3 → 51.7%, WebShop 20.7 → 41.3%, DBBench 65.6 → 70.6%, OS flat (48.6 → 49.5%).

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Paper · Code